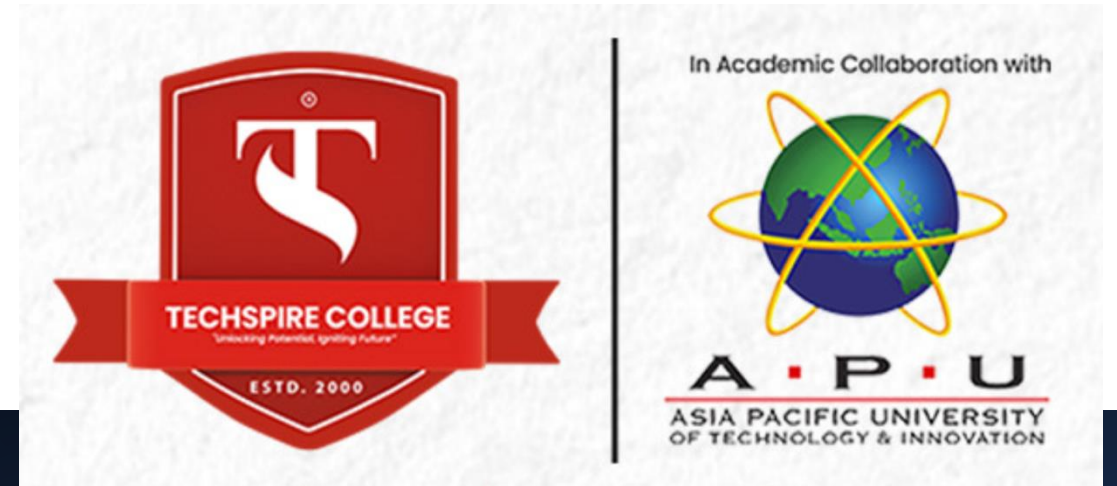


OOP in Java

Slide 6: Data Types, Comments, and Scanner

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What are data types

Data types define the type of data a variable can hold.



In Java, there are two categories:

Primitive Data Types
– Basic data types.

Reference Data Types
– Objects and arrays.

Primitive Data Types

- **Primitive Data Types in Java:**
 - **byte:** 1 byte (8 bits), range: -128 to 127.
 - **short:** 2 bytes, range: -32,768 to 32,767.
 - **int:** 4 bytes, range: -2^{31} to $2^{31}-1$.
 - **long:** 8 bytes, range: -2^{63} to $2^{63}-1$.
 - **float:** 4 bytes, used for decimals (e.g., 3.14f).
 - **double:** 8 bytes, for precise decimals (e.g., 3.141592).
 - **char:** 2 bytes, stores a single character (e.g., 'A').
 - **boolean:** 1 bit, stores true or false.

Explanation

byte: Holds small integer values. Example: `byte smallNumber = 127;`

short: A slightly larger range than byte. Example: `short mediumNumber = 32000;`

int: Used for most integer calculations. Example: `int largeNumber = 1000000;`

long: Used for very large integers. Example: `long veryLargeNumber = 100000000000L;`

float: Used for single-precision decimal numbers. Example: `float pi = 3.14f;`

double: More precision for decimal numbers. Example: `double precisePi = 3.1415926535;`

char: Holds a single character. Example: `char initial = 'A';`

boolean: Can hold true or false. Example: `boolean isJavaFun = true;`

Example

```
// Integer types
byte smallNumber = 127;           // 1 byte, range: -128 to 127
short mediumNumber = 32000;       // 2 bytes, range: -32,768 to 32,767
int largeNumber = 1000000;        // 4 bytes, range: -2^31 to 2^31-1
long veryLargeNumber = 10000000000L; // 8 bytes, use 'L' for long literals

// Floating-point types
float pi = 3.14f;                 // 4 bytes, use 'f' for float literals
double precisePi = 3.1415926535; // 8 bytes, more precision than float

// Character type
char initial = 'A';               // 2 bytes, single character, Unicode

// Boolean type
boolean isJavaFun = true;         // 1 bit, can be true or false

// Print all the variables
System.out.println("byte: " + smallNumber);
System.out.println("short: " + mediumNumber);
System.out.println("int: " + largeNumber);
System.out.println("long: " + veryLargeNumber);
System.out.println("float: " + pi);
System.out.println("double: " + precisePi);
System.out.println("char: " + initial);
System.out.println("boolean: " + isJavaFun);
```

Comments in Java

- What are Comments?
 - Comments help us understand code.
 - They are ignored by the compiler.
- Types of Comments in Java:
 - Single-line comment: Use //
 - Multi-line comment: Use /* ... */
 - Documentation comment: Use /** ... */

```
// This is a single-line comment  
int a = 10; /* This is a multi-line comment  
            used to describe the variable */  
  
// Documentation comments are used to explain methods or classes
```

Scanner for User Input

- **What is Scanner?**
 - The **Scanner** class is used to get input from the user.
 - Import the Scanner class from the java.util package.
 - `import java.util.Scanner;`

```
UserInputExample.java x
1  import java.util.Scanner; // Import the Scanner class
2
3  public class UserInputExample {
4      public static void main(String[] args) {
5          Scanner scanner = new Scanner(System.in);
6
7          // Getting input from the user
8          System.out.print("Enter your age: ");
9          int age = scanner.nextInt(); // Read integer input
10
11          System.out.print("Enter your name: ");
12          String name = scanner.next(); // Read String input
13
14          System.out.println("Hello, " + name + "! You are " + age + " years old.");
15      }
16  }
17
```

Summary

Data Types: Different kinds of values variables can hold.

Comments: How to write single-line, multi-line, and documentation comments.

Scanner: How to take user input and use it in your program.